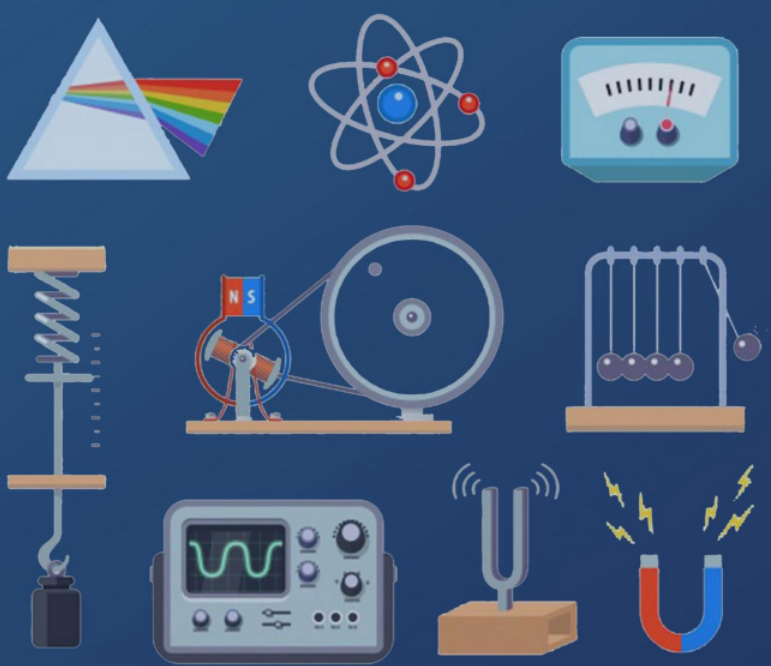
PHYSICS

Kinematics 2D



SHEET

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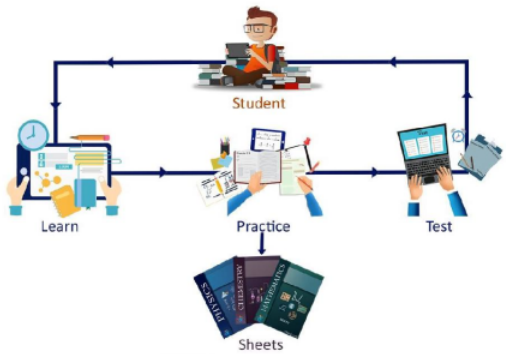
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topic	Ex - 1	Ex - 1A	Ex - 2	Ex - 2A
1	Kinematics 2D-Introduction	1, 2	7, 8	1	
2	Projectile - Part-1	3, 4			
3	Projectile - Part-2	5, 6	1-4	2, 3	10
4	Projectile - Part-3	7, 8, 9		4, 5	1, 2, 11
5	Projectile on incline		9	6	7-9
6	Equation of Trajectory		5	7, 8	12
7	Relative Motion-2D	10, 11	10	9, 10	
8	Condition of Collision	12	6		3-5, 13
9	River Boat & Rain Man	13, 14, 15			
10	After Chapter Completion				6

Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Kinematics 2D - Introduction

Q1. The coordinates of a moving particle at any time t are $x = at^2$, $y = bt^2$. The speed of particle at time t is :-

(A) $2t\sqrt{a^2+b^2}$ (B) $t\sqrt{a^2+b^2}$

(C) $2t\sqrt{a^2-b^2}$ (D) $2t\sqrt{a+b}$

Q2. A particle has initial velocity of $(\hat{i}+\hat{j})$m/s and an acceleration of $(\hat{i}+\hat{j})$ m/s^2. Its speed after 10s will be :-

(A) 11 m/s (B) 22 m/s

(C) $2\sqrt{11}$ m/s (D) $11\sqrt{2}$ m/s

Projectile Motion

Q3. Horizontal range of a projectile depends on:-

(A) Only on vertical component of velocity

(B) Only on Horizontal component of velocity

(C) Both on Horizontal and vertical component of velocity

(D) Neither horizontal nor vertical component of velocity

Q4. At the top of the trajectory of a ground to ground projectile the direction of its velocity and acceleration are :-

(A) Parallel to each other

(B) Inclined at an angle of 45° to the horizontal

(C) Perpendicular to each other

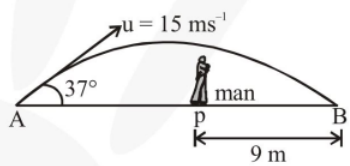
(D) None of the above statement is correct

Q5. A particle is projected with an initial velocity of 50 m/s at an angle of 60° with the horizontal then the time after which the initial velocity of the particle becomes perpendicular to the final velocity :

(A) 10 s (B) $\frac{10}{\sqrt{3}}$s

(C) $10\sqrt{3}$s (D) 12 s

Q6. A ball is hit by a batsman at an angle of 37° as shown in figure. The man standing at P should run at what minimum velocity so that he catches the ball before it strikes the ground. Assume that height of man is negligible in comparison to maximum height of projectile :-



(A) 3 ms^{-1} (B) 5 ms^{-1}

(C) 9 ms^{-1} (D) 12 ms^{-1}

Q7. When a particle is projected at an angle of 15° from horizontal its horizontal range is R. What should be the other angle from horizontal so that the horizontal range remains same if projected with same speed:-

(A) 85° (B) 45°

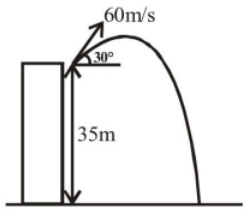
(C) 75° (D) 90°

Q8. The horizontal range of a ground to ground projectile is $2\sqrt{3}$ times its maximum height. Find angle of projection :-

(A) $\tan^{-1}\left(\frac{\sqrt{2}}{3}\right)$ (B) $\sin^{-1}\left(\frac{2}{3}\right)$

(C) $\tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$ (D) $\cot^{-1}\left(\sqrt{\frac{2}{3}}\right)$

Q9. Time of flight for the projectile shown in figure will be -



(A) 5sec (B) 6sec
(C) 7sec (D) 8sec

Relative Motion

Q10. Two particles 1 and 2 are moving with constant velocities of $\vec{v}_1 = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{v}_2 = -\hat{i} + 2\hat{j} + \hat{k}$ respectively then the velocity of particle 1 with respect to particle 2 is :

(A) $\vec{v}_{12} = 3\hat{i} + 3\hat{j} + 2\hat{k}$

(B) $\vec{v}_{12} = \hat{i} + \hat{j}$

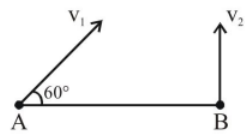
(C) $\vec{v}_{12} = \hat{i} + \hat{j} + 2\hat{k}$

(D) $\vec{v}_{12} = 3\hat{i} - 3\hat{j}$

Q11. A flag is mounted on a car moving due North with velocity of 20 km/hr. Strong winds are blowing due East with velocity of 20 km/hr. The flag will point in direction

(A) East (B) North-East
(C) South-East (D) South-West

Q12. A body is projected with velocity v_1 from point A as shown. At the same time another body is projected with velocity v_2 from point B. What should be the value of $\frac{v_1}{v_2}$ for the two bodies to collide with each-other ?



(A) $\frac{2}{\sqrt{3}}$ (B) $\frac{1}{2}$

(C) $\frac{\sqrt{3}}{\sqrt{4}}$ (D) 2

River Boat & Rain Man

Q13. A man can swim in still water with velocity of 4 m/s. River is flowing with velocity of 3m/s. If width of river is 60 m then minimum time required to cross the river will be :-

(A) 12 s (B) 15 s
(C) 20 s (D) 30 s

Q14. Wind is blowing in the north direction at speed of 2 m/s which causes the rain to fall at some angle with the vertical. With what velocity should a cyclist drive so that the rain appears vertical to him:

(A) 2 m/s south
(B) 2 m/s north
(C) 4 m/s west
(D) 4 m/s south

Q15. A boat having a speed of 5 km/hr in still water, crosses a river of width 1 km along the shortest possible path in 15 minutes. The speed of the river in Km/hr.

(A) 1 (B) 3
(C) 4 (D) $\sqrt{41}$

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Suppose a player hits several baseballs. Which baseball will be in the air for the longest time?

(A) The one with the farthest range.

(B) The one which reaches maximum height.

(C) The one with the greatest initial vertical component of velocity.

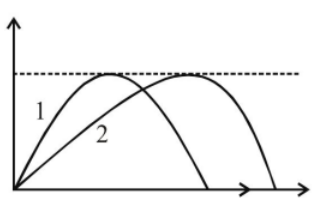
(D) The one leaving the bat at 45° with respect to the ground.

Q2. Two particles are projected from ground with same initial velocities at an angle 30° and 60° with horizontal. If R_1 and R_2 be their horizontal ranges, H_1 and H_2 their maximum heights and T_1 and T_2 are the time of flight, then :-

(A) $\frac{H_1}{R_1} > \frac{H_2}{R_2}$ (B) $\frac{H_1}{R_1} < \frac{H_2}{R_2}$

(C) $\frac{H_1}{T_1} > \frac{H_2}{T_2}$ (D) None

Q3. Trajectories of 2 projectiles are shown. Let u_1 and u_2 are velocities of their projection and T_1, T_2 are the times of flight. Then



(A) $T_2 = T_1$ (B) $T_2 > T_1$

(C) $u_1 > u_2$ (D) $u_1 < u_2$

Q4. A particle is projected from the ground with velocity u at angle θ with horizontal. The horizontal range, maximum height and time of flight are R, H and T respectively.

They are given by $R = \frac{u^2 \sin 2\theta}{g}$,

$H = \frac{u^2 \sin^2 \theta}{2g}$ and $T = \frac{2u \sin \theta}{g}$.

Now keeping u fixed, θ is varied from 30° to 60°, then the INCORRECT statement(s) are

(A) R will first increase then decrease, H will increase and T will decrease.

(B) R will first increase then decrease while H and T both will increase.

(C) R will decrease while H and T both will increase.

(D) R will increase while H and T both will also increase.

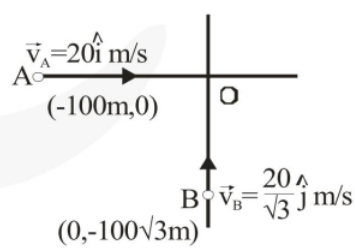
Q5. The equation of a projectile is $y = \sqrt{3}x - \frac{gx^2}{2}$.

The angle of projection is :-

(A) 30° (B) 60°

(C) 45° (D) None

Q6. Positions of two vehicles A and B with reference to origin O and their velocities are as shown.



(A) they will collide

(B) distance of closest approach is 100 m.

(C) their relative speed is $\frac{40}{\sqrt{3}}$ m/s

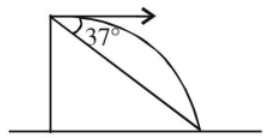
(D) their relative speed is $\frac{20}{\sqrt{3}}$ m/s

SUBJECTIVE TYPE

Q7. The position of a particle is given by $\vec{r} = (3t\,\hat{i} - 2t^2\,\hat{j} + 4\hat{k})$m.
where t is in seconds and the coefficients have the proper units for r to be in metres.
Find the v and a of the particle.

Q8. The vertical height y and horizontal distance x of a projectile on a certain planet are given by x = (3t)m, y = $(4t - 6t^2)$ m where t is in seconds. Find the speed of projection (in m/s).

Q9. A ball is thrown horizontally from a cliff such that it strikes ground after 5 s. The line of sight from the point of projection to the point of landing makes an angle of 37° with the horizontal.
What is the initial velocity of projection?



Q10. A glass wind-screen of adjustable inclination is mounted on a car. The car moves horizontally with a speed of 6 m/s. At what angle α with the vertical should the wind screen be adjusted so that the rain drops falling vertically with 2 m/s strike the wind screen perpendicularly?

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Position vector of a particle with respect to projection point is $\vec{r} = 3t\hat{i} + (4t - 5t^2)\hat{j}$. Take horizontal direction on x-axis and vertical direction on y-axis. Then find out angle of projection :-
(A) 53°
(B) 30°
(C) 60°
(D) Can't be determine

Q2. A particle is projected at an angle of 60° from horizontal with velocity of 50 m/s, it will make an angle of 30° from horizontal after time :-
(A) $\frac{5}{\sqrt{3}}$ s (B) $5\sqrt{3}$ s
(C) $\frac{10}{\sqrt{3}}$ s (D) $10\sqrt{3}$ s

Q3. A particle is projected from a horizontal plane (x-z plane) such that its velocity vector at time t is given by $\vec{V} = a\hat{i} + (b - ct)\hat{j}$. Its range on the horizontal plane is given by -
(A) $\frac{ba}{c}$ (B) $\frac{2ba}{c}$
(C) $\frac{3ba}{c}$ (D) None

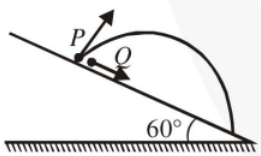
Q4. A cannon ball a has range R on a horizontal plane. If h and h' are the greatest heights in the two paths for which this is possible, then-
(A) $R = 4\sqrt{(hh')}$ (B) $R = \frac{4h}{h'}$
(C) $R = 4\,hh'$ (D) $R = \sqrt{hh'}$

Q5. A particle is projected from ground towards a vertical wall 80 m away at an angle of 37° with horizontal with initial velocity of 50 m/s. After its collision with wall & then once with ground find at what distance in meter from wall will it strike the ground. The component of velocity normal to the surface becomes half after collision with each surface.

(A) 100 (B) 140

(C) 150 (D) 120

Q6. A particle P is projected from a point on the surface of smooth inclined plane (see figure). Simultaneously another particle Q is released on the smooth inclined plane from the same position. P and Q collide after $t = 4$ s. The speed of projection of P is :-



(A) 5 m/s (B) 10 m/s

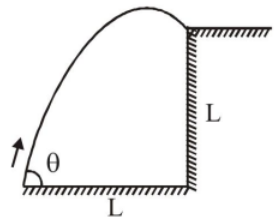
(C) 15 m/s (D) 20 m/s

Q7. If $y = x - \frac{1}{2}x^2$ is equation of trajectory then time of flight is (g is acceleration of objects).

(A) $\sqrt{2g}$ (B) $\frac{2}{\sqrt{g}}$

(C) g (D) $2.7\sqrt{g}$

Q8. A ball is thrown at an angle θ up to the top of a cliff of height L, from a point at a distance L from the base, as shown in figure. Assuming that one of the following quantities is the initial speed required to make the ball hit right at the edge of the cliff, which one is it :-



(A) $\sqrt{\frac{gL}{2(\tan\theta - 1)}}$

(B) $\frac{1}{\cos\theta}\sqrt{\frac{gL}{2(\tan\theta - 1)}}$

(C) $\frac{1}{\cos\theta}\sqrt{\frac{gL}{2(\tan\theta + 1)}}$

(D) $\sqrt{\frac{gL\tan\theta}{2(\tan\theta + 1)}}$

Q9. A ball is dropped from a building of height 45m. Simultaneously another ball is thrown up with a speed 40 m/s. Find the relative speed of the balls as a function of time till both balls are in air.

(A) remains 40 m/s

(B) Zero

(C) (40–10t) m/s

(D) First increases then decreases

Q10. Three ships A, B & C are in motion. Ship A moves relative to B is with speed v towards North-East. Ship B moves relative to C with speed v towards the North-West. Then relative to A, C will be moving towards.

(A) North (B) South

(C) East (D) West

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Two particles *A* and B projected along different directions from the same point *P* on the ground with the same velocity of 70 m/s in the same vertical plane . They hit the ground at the same point *Q* such that *PQ* = 480 m. Then [g = 9.8 m/s^2]

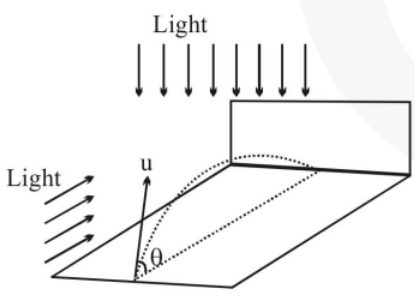
(A) Ratio of their times of flight is 4 : 5

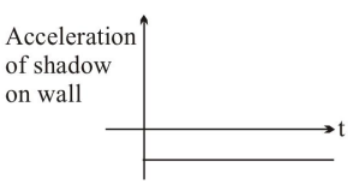
(B) Ratio of their maximum heights is 9 : 16

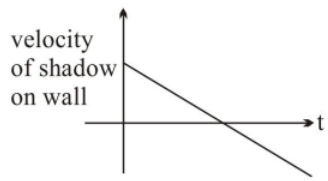
(C) Ratio of their minimum speeds during flights is 4 : 3

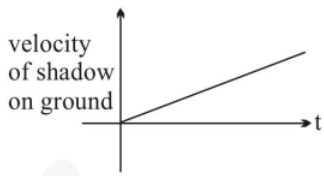
(D) The bisector of the angle between their directions of projection makes 45° with horizontal

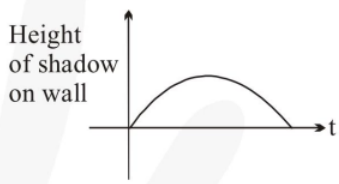
Q2. A projectile is projected as shown in figure. A proper light arrangement makes a shadow on the wall as well as on the floor ? Which of the following graphs is/are correct. Taking upward direction as positive.



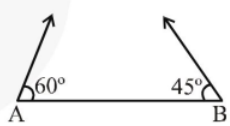
(A)


(B)


(C)


(D)


Q3. Two particles of equal identical mass are projected with same velocity v_0 at angle 60° and 45° simultaneously from the ground as shown in figure. If both the p articles are projected in same plane. Then for the time when both particles are in air :-



(A) Particle B appears to move in straight line as seen by A

(B) Magnitude of velocity of A relative to B increases continuously.

(C) Acceleration of A relative to B is zero during the motion.

(D) Acceleration of A relative to B is constant during their motion.

Q4. Two particles A and B are located in x-y plane at points (0, 0) and (0, 4). They simultaneously start moving with velocities $\vec{V}_A = 2\hat{j}\,m/s$ and $\vec{V}_B = -2\hat{i}\,m/s$.

Select the correct option.

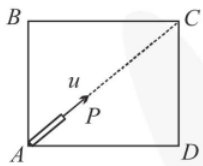
(A) The distance between them is constant.

(B) The distance between them first decreases and then increases.

(C) The shortest distance between them is $2\sqrt{2}$ m.

(D) Time after which they are at minimum distance is 1 sec.

Q5. A large rectangular box moves vertically downward with an acceleration a. A toy gun fixed at A and aimed towards C fires a particle P



(A) P will hit C if $a = g$

(B) P will hit the roof BC, if $a > g$

(C) P will hit the wall CD if $a < g$

(D) In all cases P will hit C.

Q6. A particle moves in the xy-plane and at time t is at the point $(t^2,\ t^3 - 2t)$. Then

(A) At $t = 2/3$ s, directions of velocity and acceleration are perpendicular

(B) At $t = 0$, directions of velocity and acceleration are perpendicular

(C) At $t = \sqrt{2/3}$ s, particle is moving parallel to x-axis

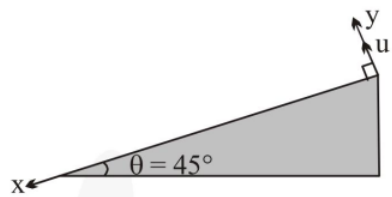
(D) Acceleration of the particle when it is at point (4, 4) is $2\hat{i} + 12\hat{j}$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # (Ques. 7 to 9)

An inclined plane makes an angle $\theta = 45°$ with horizontal. A stone is projected normally from the inclined plane, with speed u m/s at t = 0 (as shown in figure). x and y axis are drawn from point of projection along and normal to inclined plane as shown. The length of incline is sufficient for stone to land on it and neglect air friction.



Q7. The instant of time at which velocity of stone is parallel to x-axis

(A) $\frac{2\sqrt{2}\,u}{g}$ (B) $\frac{2u}{g}$

(C) $\frac{\sqrt{2}u}{g}$ (D) $\frac{u}{\sqrt{2}\,g}$

Q8. The instant of time at which velocity of stone makes an angle $\theta = 45°$ with positive x-axis i.e. when velocity become horizontal.

(A) $\frac{2\sqrt{2}\,u}{g}$ (B) $\frac{2u}{g}$

(C) $\frac{\sqrt{2}u}{g}$ (D) $\frac{u}{\sqrt{2}\,g}$

Q9. The instant of time at which (starting from t = 0) component of displacement along x-axis is half the range on inclined plane is-

(A) $\frac{2\sqrt{2}\,u}{g}$ (B) $\frac{2u}{g}$

(C) $\frac{\sqrt{2}u}{g}$ (D) $\frac{u}{\sqrt{2}\,g}$

MATCH THE COLUMN TYPE

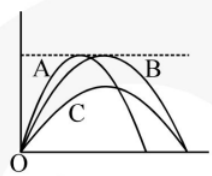
*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q10. A body is projected with velocity v at an angle of projection θ with the vertical then,

(A) Vertical component of velocity	(P) Maximum at highest point
(B) Kinetic energy of body	(Q) Minimum at highest point (Not Zero)
(C) Shorter angle with vertical	(R) At highest pt. is zero
(D) Horizontal component of velocity	(S) Remains unchanged

(A) (A) $\rightarrow$ R; (B) $\rightarrow$ Q;(C) $\rightarrow$ P;(D) $\rightarrow$ S
(B) (A) $\rightarrow$ R; (B) $\rightarrow$ Q;(C) $\rightarrow$ S;(D) $\rightarrow$ P
(C) (A) $\rightarrow$ Q; (B) $\rightarrow$ R;(C) $\rightarrow$ P;(D) $\rightarrow$ S
(D) (A) $\rightarrow$ Q; (B) $\rightarrow$ R;(C) $\rightarrow$ S;(D) $\rightarrow$ P

Q11. Trajectories are shown in figure for three kicked footballs. Initial vertical & horizontal velocity components are u_y and u_x respectively. Ignoring air resistance, choose the correct statement from **Column II** for the value of variable in **Column I**.



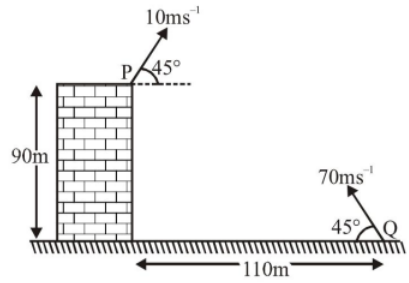
Column I	**Column II**
(A) time of flight	(P) greatest for A only
(B) u_y/u_x	(Q) greatest for C only
(C) u_x	(R) equal for A and B
(D) $u_x u_y$	(S) equal for B and C

Q12. Trajectory of particle in a projectile motion is given as $y = x - \frac{x^2}{80}$. Here, x and y are in meters. For this projectile motion, match the following with g=10 m/s^2.

Column–I	**Column–II**
(A) Angle of projection (in degrees)	(P) 20
(B) Angle of velocity with horizontal after 4s (in degrees)	(Q) 80
(C) Maximum height (in metres)	(R) 45
(D) Horizontal range (in metres)	(S) 30
	(T) 60

SUBJECTIVE TYPE

Q13. Two particles P and Q are launched simultaneously as shown in figure. Find the minimum distance between particles in meters.



Answer Key

Ex-1									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
A	D	C	C	B	B	C	C	C	D
11	**12**	**13**	**14**	**15**					
C	A	B	B	B					

Ex - 1A

1. B, C **2.** B **3.** A, D **4.** A, C, D **5.** B

6. B, C **7.** $\vec{V}(t)=\left(3\hat{i}-4t\hat{j}\right)m/s, \vec{a}(t)=-4\hat{j}\,m/s^2$ **8.** 5

9. $\frac{100}{3}m/s$ **10.** $\tan^{-1}\left(\frac{1}{3}\right)$

Ex-2									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
A	A	B	A	B	B	B	B	A	B

Ex - 2A

1. B, C, D **2.** A, B, D **3.** A, C, D **4.** B, C, D **5.** AB

6. A, B, C, D **7.** C **8.** D **9.** B **10.** A

11. (A)-R, (B)-P, (C)-Q, (D)-S **12.** (A)-R, (B)-R, (C)-P, (D)-Q **13.** 6

-: Important Notes & Formulas :-